

Feature Extraction and Classification of EEG Signal Based on Deep Learning

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Abstract—In order to remove the noises mixed in EEG signals and improve the classification accuracy of motor imagery EEG signals, an EEG processing method based on EMD+ICA+ deep learning is proposed in this paper. First, both independent component analysis (ICA) and empirical mode decomposition (EMD) are used to remove noise interferences such as electrooculogram (EOG) and electrocardiogram (ECG) mixed in EEG signals. This method can retain useful EEG information while denoising. It can provide good conditions for further feature extraction and pattern recognition. The second step is to use the time-frequency analysis method to combine the time, frequency and position information extracted from the EEG signals to extract the time-frequency diagram of the EEG signal. The time-frequency diagram is used to characterize the complex information of the EEG signals. The third step is to take time-frequency graph as the input of a 2-layer convolutional neural network (CNN) with a two-dimensional convolution and maximum pooling layer. In the neural network, the generalization of the model is improved by introducing batch normalization (BN), and the rectified unit (ReLU) is used in the convolutional layer to accelerate the training. The feature extraction and classification steps are combined into one process for feature extraction and classification. Finally, the classification accuracy are used in the experiment to evaluate the performance of the method. The experimental results show that the neural network classification method proposed in this paper has high accuracy and reliability.

Keywords- BCI; CNN; EEG; Time-frequency diagram ; Wavelet transform; ICA; EMD

I. INTRODUCTION

In recent years, the application of brain-computer interfaces has been extensively studied. Research is mainly concentrated in the field of biomedical engineering. The main idea of brain-computer interface (BCI) is to use thinking to control computers or robotic devices without any physical intervention.

Various types of EEG signals are used as the control signals of the brain-computer interface in the research of brain-computer interface. Among the control signals, P300 evoked potential, steady-state visual evoked potential (SSVEP) and motor imaging (MI) are the most common signals. This paper treats brain potentials related to motor imaging tasks as control signals.

The EEG signal is a non-stationary signal. It also has the characteristics of weak, low signal-to-noise ratio, and strong randomness. These characteristics make our collected signals contain some interference signals. These interference signals will have an undesirable effect on the EEG analysis process. Therefore, this paper uses a denoising algorithm based on the combination of independent component

analysis and empirical mode decomposition to preprocess the original EEG signals.

Deep convolutional neural networks have been proven to capture the intrinsic characteristics of brain electrical signals in the field of computer vision. CNN introduces a convolutional layer. The weight sharing and sparse connection characteristics of the convolutional layer significantly reduce the computational burden and improve the model's robustness to signal variance. Another feature of CNN is that it can extract the hierarchical features of the original data without too much preprocessing. Using this advantage, CNN can combine feature extraction and recognition into one process.

II. DATA SET

This experiment uses the 4th International Brain-Computer Interface Competition-Goetz Data Set 2a (BCI Computer IV-Data Set 2a).

The subject sat comfortably in the armchair in front of the computer screen. At the beginning of the experiment ($t=0s$), a gaze cross appeared on the black screen, and a short audible warning sound was also given. Two seconds later ($t=2s$), an arrow-shaped prompt appeared on the screen. The prompt points to the left, right, up or down respectively (corresponding to one of the four categories of left hand, right hand, foot or tongue). The prompt stays on the screen for 1.25 seconds. The subjects performed the suggested motor imaging tasks. Until $t=6s$, the gaze cross disappears from the screen. After a short break, the screen went black again. The time sequence is shown in Figure 1. In this experiment, 22 Ag/AgCl electrodes were used to record EEG signals, of which 3 electrodes were used to collect EOG signals. The data in all experiments have been processed with a band-pass filter of 0.5-100HZ and a notch filter of 50HZ. The sampling frequency of the whole experiment is 250HZ, and the amplitude range of EEG data is $\pm 50\mu v$.

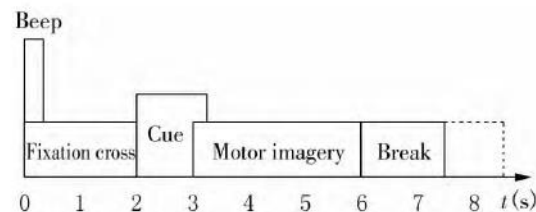


Figure 1. Time schedule of the experiment

III. DENOISING PROCESSING OF EEG SIGNALS

A. Empirical model decomposition

The Empirical Mode Decomposition (EMD) method was proposed by N.E. Huang et al. in 1998. It has great advantages for analyzing nonlinear and non-stationary signal

sequences, and at the same time has the characteristics of high signal-to-noise ratio. The core of this method is Empirical Mode Decomposition, which decomposes a complex signal into a finite number of Intrinsic Mode Functions (IMF) arranged in descending order of frequency. The decomposed IMF components contain local characteristic signals of different time scales of the original signal.

The key of EMD is to decompose a complex signal into a finite number of eigenmode functions (IMF), and each of the decomposed IMF components contains local characteristic signals of different time scales of the original signal.

This paper selects the EEG data of the first subject for EMD decomposition. Figure 2 is a schematic diagram of the EMD decomposition of the EEG data and the instantaneous frequency of the corresponding IMF. It can be seen from Figure 2 that the original EEG signal $X(t)$ is decomposed by EMD to obtain 9 IMF components, and the obtained IMF components are arranged in descending order of frequency.

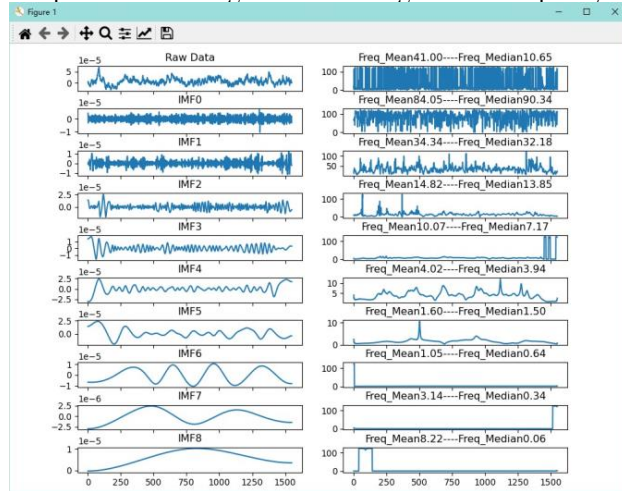


Figure 2. EMD decomposition results of the first subject's EEG data

It can be seen from the decomposition results in Figure 2 that each level of IMF represents the characteristic signals of different time scales in the original EEG signal. In each order component, IMF0 and IMF1 are high-frequency noise. It can be seen that EMD has good adaptability to the decomposition of EEG signals. The basic principle of EMD denoising is to remove the IMF component composed of high frequency noise. Therefore, when using the EMD method to denoise the EEG signal. We can achieve the denoising effect by deleting the IMF component of the high frequency part. Then we superimpose the remaining part of the IMF to get the denoised EEG signal.

B. Independent component analysis

ICA (Independent Component Analysis, ICA) is a multi-channel signal processing and data analysis method. It can separate the multi-channel observation signals formed by the aliasing of statistically independent "source" signals, and restore the original multiple independent "source" signals.

The traditional EEG noise analysis method is to directly remove the records with significant artifact characteristics from the measurement signal. However, this method can easily lead to the loss of a large amount of EEG data, especially the data of brain activity during eye movement.

This ultimately affects the results of the analysis. EEG signals and eye movement noise are generated by different signal sources that are independent of each other. Therefore, eye movement artifacts mixed in EEG data can be effectively separated with the help of ICA technology.

C. Denoising Processing of EEG Signal Based on EMD-ICA

The specific processing steps are as follows:

1) First, we use empirical mode decomposition (EMD) to decompose the EEG signal to obtain the intrinsic mode component (IMFS) and margin;

2) Then we take the first three-order eigenmode components to construct a virtual noise channel;

3) We use FastICA to analyze and process the signal channel and virtual noise channel 1 to realize the separation of the components, and the ICA components that meet the requirements are screened.

4) Repeat steps 2) and 3) until all IMF components are separated by ICA to eliminate noise.

After denoising the EEG signal by the above method, the interference of the noise signal can be eliminated to the greatest extent on a finer scale. At the same time, most useful EEG signals are retained, so as to achieve the purpose of denoising EEG signals. However, the denoising method based on EMD-ICA has shortcomings such as low computational efficiency and end effect, which needs to be improved. The EEG denoising process based on the above method is shown in Figure 3.

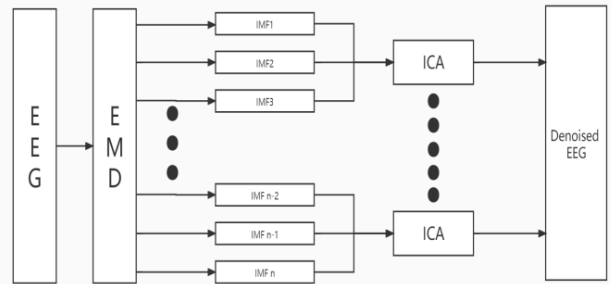


Figure 3. EMD-ICA denoising flow chart

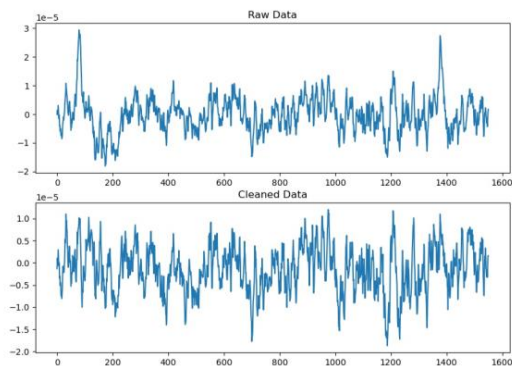


Figure 4. Experimental results of EMD-ICA method

This paper selects the first subject's EEG data for EMD-ICA decomposition. The experimental results are shown in Figure 4. The original EEG signal is not smooth and mixed with various noises. After the original signal is decomposed by EMD-ICA, the result obtained is relatively ideal. The denoising signal obtained by this method is more in line with the natural trend of the original data. It can be seen that the EMD-ICA algorithm not only accurately eliminates the

artifact components, but also retains the local characteristics of the original EEG.

The EMD-ICA method is effective in eliminating information redundancy and improving the amount of independence between intrinsic modal functions. Using the denoising algorithm based on EMD-ICA can eliminate the noise signal on a finer scale, while retaining the local characteristics of the EEG signal.

IV. WAVELET TRANSFORM

Wavelet transform is a time-frequency analysis method proposed by J. Morlet when analyzing geophysical survey materials. The theoretical basis is the Fourier transform. With the development of wavelet analysis theory, it even has a higher status than Fourier transform in signal processing. Wavelet transform is regarded as a milestone development since the Fourier transform was proposed. In principle, wavelet analysis can handle almost all problems that can be solved by Fourier transform.

Continuous Wavelet Transform (Continue Wavelet Transform) is to project a one-dimensional signal onto a two-dimensional time-scale plane.

The diversification of the mother wavelet function is one of the main advantages of wavelet transform. Commonly used mother wavelet functions include Haar wavelet, Morlet wavelet, Mexican Hat (Mexh) wavelet and Daubechies (DbN) wavelet.

The left-hand and right-hand motor MI tasks lead to ERD and ERS on the left and right sides of the motor cortex respectively, which affect the EEG signals of the C4 and C3 electrodes. Figure 5 and Figure 6 show the wavelet time-frequency images of the C3 and C4 channels when performing the left-hand motion imaging task. It can be seen from the figure that 3 seconds after the start of the experiment, the energy of the C4 channel dropped significantly and recovered after a period of time. This is the occurrence of ERD phenomenon. However, the energy of the C3 channel remains at a high level instead of decreasing, which is called the ERS phenomenon. Therefore, the images of C3 and C4 are combined into the final input image, as shown in Figure 7.

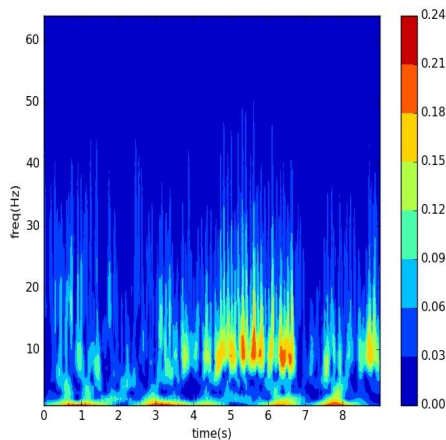


Figure 5. Wavelet time-frequency image of C3

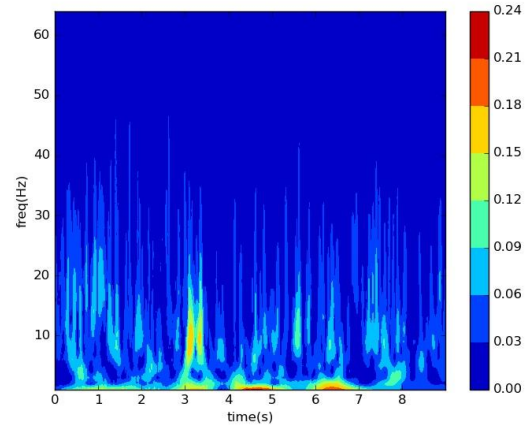


Figure 6. Wavelet time-frequency image of C4

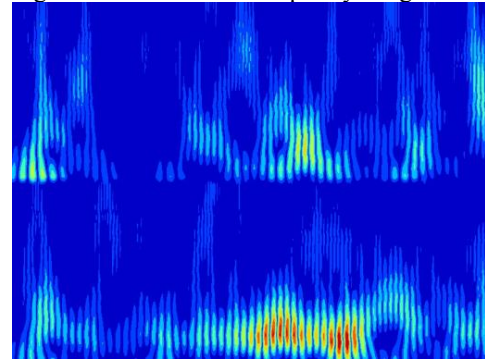


Figure 7. The final combined input image of C3 and C4

V. CONVOLUTIONAL NEURAL NETWORK

Convolutional neural networks usually consist of multiple convolutional layers and pooling layers. A typical CNN network has the following levels: input layer, convolutional layer, pooling layer, fully connected layer and output layer. In the convolutional layer, the convolution kernel is used to perform convolution filtering operations on the input data, which has the characteristics of local linkage and weight sharing. Compared with the fully connected layer, the convolutional layer can extract the local information of the input data and reduce the number of parameters. In the pooling layer, the maximum pooling or average pooling method is usually used to reduce the dimensionality of features, which reduces the complexity of network calculation.

This paper alternately calculates the convolutional layer and the pooling layer to reduce the size of the feature map and increase the number of channels. The output layer uses the softmax classification function. This function can select the item with a higher probability as the classification result to perform left- and right-handed binary classification judgment on the input. It has the advantage of convenient calculation of the back propagation process.

This paper designs a smaller 2-layer convolutional neural network model. Its structure is shown in Figure 5-1. It includes 2 convolutional layers, 2 pooling layers, 2 fully connected layers and 1 softmax output layer.

In this paper, the size of the input image is uniformly adjusted to 64*64. The kernel size is 4*4 in the first convolutional layer and 2*2 in the second convolutional layer. The number of kernels in the first convolutional layer is 8, and the number of kernels in the second convolutional layer is 16. In this paper, the neural network chooses 8 batch size and 300 epochs. In the training of the neural network, the gradient descent algorithm is used iteratively. The ADAM algorithm is used to obtain the adaptive learning rate of each parameter. In addition, this paper innovatively introduces Batch Normalization (BN) to alleviate over-fitting. And Rectified Unit (ReLU) is used in the convolutional layer to accelerate training.

VI. EXPERIMENTAL RESULTS

In order to quantitatively evaluate the algorithm in this paper, three different methods of ICA, ICA-WT and EMD-ICA were preprocessed on the original signal. The comparison results are shown in Table 1. It can be seen that the denoising effect of ICA alone is the worst. Although the denoising effect of ICA-WT is better than that of ICA alone, the final classification accuracy is still not as good as the EMD-ICA algorithm. Using the EMD-ICA algorithm proposed in this paper improves the signal-to-noise ratio of the signal. At the same time, the algorithm retains the effective components of the signal well. Therefore, it has a more ideal noise reduction effect. Finally, the classification accuracy is effectively improved.

Table 1 Comparison of classification accuracy of ICA, ICA-WT and EMD-ICA denoising algorithms

Methods	Accuracy(%)		
	Worst	Best	Mean
ICA	63.55	72.30	68.21
ICA-WT	78.70	88.49	82.26
EMD-ICA	83.53	90.16	86.45

In this paper, wavelet transform is used for time-frequency analysis of EEG signals. The choice of mother wavelet is flexible, which is the advantage of wavelet transform. It is also the link that wavelet transform needs to pay attention to. From the experimental results, the results obtained by different mother wavelets are very different. This paper selects different mother wavelets for comparison, and the results are shown in Table 2. It can be seen that the classification result based on the Morlet wavelet transform is relatively good, and the calculation cost is lower than that of the Mexh wavelet function.

Table 2 Classification results of different mother wavelet functions

Methods	Accuracy(%)			Average Times(s)
	Worst	Best	Mean	
Haar	73.53	78.27	75.15	76.37
Morlet	84.12	89.49	87.26	79.53
Mexh	81.44	85.16	82.40	87.09
db4	76.39	82.36	79.71	80.46

This paper uses the neural network described above to classify the data of 9 subjects. It also applies support vector machines (SVM) and stacked autoencoders (SAE) to the same input image. It can be seen from Table 3 that the

average accuracy of the CNN method proposed in this paper is higher than that of SVM and SAE under the same input data.

VII. CONCLUSIONS

Aiming at the problem of motor imagery EEG signal classification, this paper proposes a method of EEG signal classification based on time-frequency images and convolutional networks to classify EEG motor imagery signals into two types: right-hand movement and left-hand movement. In this method, the pretreatment method combining EMD and ICA is first used. The EMD-ICA algorithm provides a good premise for feature extraction and pattern recognition, and improves the accuracy of the final result. Then we use the time, frequency and location information of the EEG signal to design a new input form. This paper uses wavelet transform to extract the time-frequency images of C3 and C4 channels and combines them as input for CNN network. The filters in the convolutional layer of CNN are trained to learn the position and frequency of the activation mode. At the same time, their position on the time axis is partially unchanged due to the maximum pooling. The input in the form of an image also preserves the relative position between the channels. In order to obtain a stable and reliable model, K-fold cross-validation is used to evaluate the performance of the classifier. Experiments show that the convolutional neural network method can achieve better results than traditional methods.

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